Machine Learning

18-05-26

Adaboost Regression :

X_1 : [d_1 d_2 d_3 d_4 d_5]

	x_1	x_2	y	w_1		$\hat{y}_1$
d_1	1	1	1	$1/5 = 0.2$	train model m_1 DTree regression on X_1	1
d_2	1	1	2	0.2		2.2
d_3	1		4	0.2		4.1
d_4	1	1	6	0.2		6
d_5	1	1	7	0.2		7.5

	x_1	x_2	y	w_1	$\hat{y}_1$	$L_1' = \lvert y - \hat{y}_1 \rvert$	$L_1 = \frac{L_1'}{\max(L_1')}$
d_1		1		0.2	1	0	$0/0.5 = 0$
d_2		2		0.2	2.2	0.2	$0.2/0.5 = 0.4$
d_3		4		0.2	4.1	0.1	$0.1/0.5 = 0.2$
d_4		6		0.2	6	0	$0/0.5 = 0$
d_5		7		0.2	7.5	0.5	$0.5/0.5 = 1$

from L_1 values we can see that

d_1 and d_4 → no mistake

d_3 → small mistake

d_2 and d_5 → large mistake

$$\text{Total Error} = \sum_{i=1}^{5} L_i w_i$$

$$= 0 \times 0.2 + 0.4 \times 0.2 + 0.2 \times 0.2 + 0 \times 0.2 + 1 \times 0.2$$

$$TE_1 = 0.32$$

model say factor = $\alpha_1 = \frac{1}{2} \log\left(\frac{1-TE_1}{TE_1}\right)$

$$\alpha_1 = \frac{1}{2} \log\left(\frac{1}{\beta_1}\right)$$

where $\beta_1 = \frac{TE_1}{1-TE_1}$

$$\alpha_1 = \frac{1}{2} \log\left(\frac{1}{\beta_1}\right) = \frac{1}{2} \log\left(\frac{1-TE_1}{TE_1}\right) = \frac{1}{2} \log\left(\frac{1-0.32}{0.32}\right) = 0.16$$

$$\alpha_1 = 0.16 \qquad \beta_1 = \frac{0.32}{1-0.32} = 0.47$$

we want next model to focus more on datapoints where m1 made mistakes i.e. d_3 d_2 and d_5

	x_1	x_2	y	w_1	L_1	$1-L_1$	$w_2' = w_1 \times \beta_1^{1-L_1}$	$w_2 = w_2'/\sum w_2'$
d_1	1	0.2	0	1	$0.2 \times 0.47^1 = 0.094$	$0.094/0.628 = 0.15$
d_2	2	0.2	0.4	0.6	$0.2 \times 0.47^{0.6} = 0.13$	0.2
d_3	4	0.2	0.2	0.8	$0.2 \times 0.47^{0.8} = 0.11$	0.175
d_4	6	0.2	0	1	$0.2 \times 0.47^1 = 0.094$	0.15
d_5	7	0.2	1	0	$0.2 \times 0.47 = 0.2$	0.32
					$\sum w_2' = 0.628$	$\sum w_2 \approx 1$

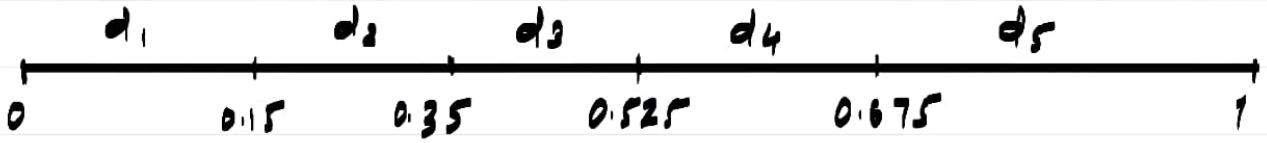
$$w_2' = w_1 \times \beta_1^{1-L_1}$$

$$w_2 = w_2' / \sum w_2'$$

w_2 values are higher for d_2 d_3 d_5 (model m_1 mistakes) compared to d_1 and d_4.

create X_2 by weighted sampling using w_2

	w_2	cummulative w_2
d_1	0.15	0.15
d_2	0.2	0.15+0.2 = 0.35
d_3	0.175	0.35+0.175 = 0.525
d_4	0.15	0.525+ 0.15 = 0.675
d_5	0.32	0.675+0.32 ≈ 1



- more weightage is given to datapoints where previous model made larger errors

- more of these datapoints with larger errors will be selected due to weighted sampling

- more of these datapoints in X_2 means model will focus more on mistakes of model m_1

$X_2 = [d_5\ d_1\ d_3\ d_1\ d_5]$

	x_1	x_2	y	w_2		$\hat{y}_2$	L_2	w_3
d_5			7	0.2	} train DTree regr model m_2 on X_2	7		
d_2			2	0.2		3		
d_3			4	0.2		3		
d_1			1	0.2		1		
d_5			7	0.2		7		

TE_2 α_2 β_2 $w_3' = w_2 \times \beta_2^{(1-L_2)}$

$w_3 = w_3' / \sum w_3'$

- generate new dataset X_3 using weighted sampling over w_3

- train next model m_3 on X_3

- continue untill the stopping criteria
  - stop if decided number of weak learners are trained
  - stop if errors are not reducing further.

Lets say we train 4 models in this series



new datapoint x



$$\text{final prediction} = \frac{\alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3 + \alpha_4 p_4}{\alpha_1 + \alpha_1 + \alpha_3 + \alpha_4}$$

model having higher α will have more impact on final prediction

Gradient Boosting Model : GBM

- used for regression and classification
- DTrees in GBM are trained based on gradients and residuals

GBM Regression

$X_1 \rightarrow [d_1\ d_2\ d_3\ d_4\ d_5]$

	x_1	x_2	y	$\hat{y}_0$	residuals $R_1 = y - \hat{y}_0$
d_1	1	1	20	12	20-12 = 8
d_2	1		15	12	15-12 = 3
d_3		1	10	12	10-12 = -2
d_4	1		8	12	8-12 = -4
d_5			7	12	7-12 = -5

$$\bar{y} = \hat{y}_0 = \frac{\sum y}{n} = \frac{60}{5} = 12$$

we will train the DTree regression model m_1 with x_1 and x_2 as features and R_1 as target

$$\text{error} = L = MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2$$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial L}{\partial z}\cdot\frac{\partial z}{\partial \hat{y}} \qquad z = y - \hat{y}$$

$$= \frac{1}{n}\frac{\partial z^2}{\partial z}\cdot\frac{\partial (y-\hat{y})}{\partial \hat{y}}$$

$$= \frac{2}{n} z \times (0-1) = -\frac{2}{n} z = \frac{-2}{n}(y-\hat{y})$$

residual

$$= -2/n\ R$$

model m_1 uses → x_1, x_2 as features
R_1 as target

model m_1 predictions → $\hat{R}_1$

	x_1	x_2	y	$\hat{y}_0$	R_1	$\hat{R}_1$	$\hat{y}_1 = \hat{y}_0 + \alpha \hat{R}_1$ with $\alpha = 0.5$
d_1		20	12	8	5.5		$12 + 0.5 \times 5.5 = 14.75$
d_2		15	12	3	5.5		$12 + 0.5 \times 5.5 = 14.75$
d_3		10	12	-2	-3.5		$12 + 0.5 \times -3.5 = 10.25$
d_4		8	12	-4	-4		$12 + 0.5 \times -4 = 10$
d_5		7	12	-5	-3.5		$12 + 0.5 \times -3.5 = 10.25$

α here is learning rate hyperparameter

now we compute R_2 for training the next model

$$R_2 = y - \hat{y}_1$$

	x_1	x_2	y	$\hat{y}_0$	R_1	$\hat{R}_1$	$\hat{y}_1$	$R_2 = y - \hat{y}_1$
d_1		20	12	8	5.5		14.75	$20 - 14.75 = 5.25$
d_2		15	12	3	5.5		14.75	$15 - 14.75 = 0.25$
d_3		10	12	-2	-3.5		10.25	$10 - 10.25 = -0.25$
d_4		8	12	-4	-4		10	$8 - 10 = -2$
d_5		7	12	-5	-3.5		10.25	$7 - 10.25 = -3.25$

train next DT regression model m_2 with
x_1, x_2 as features and
R_2 as target

GBM Regression Training:

1) Decide number of DTrees / estimators to build in series

2) start with initial prediction as

$$\hat{y}_0 = \text{avg}(y) = \bar{y}$$

$\hat{y}_0$ → avg of actual target values

3) compute Residuals $R_i = y - \hat{y}_{i-1}$

$$R_1 = y - \hat{y}_0$$
$$R_2 = y - \hat{y}_1$$
$$R_3 = y - \hat{y}_2$$
$$\vdots$$

4) Train DT model m_i with original features of data and target R_i

$$X \longrightarrow \boxed{m_i} \longrightarrow \hat{R}_i$$

5) Get residual predictions $\hat{R}_i$ from model m_i

6) compute new predictions from m_i as

$$\hat{y}_i = \hat{y}_{i-1} + \alpha \hat{R}_i$$

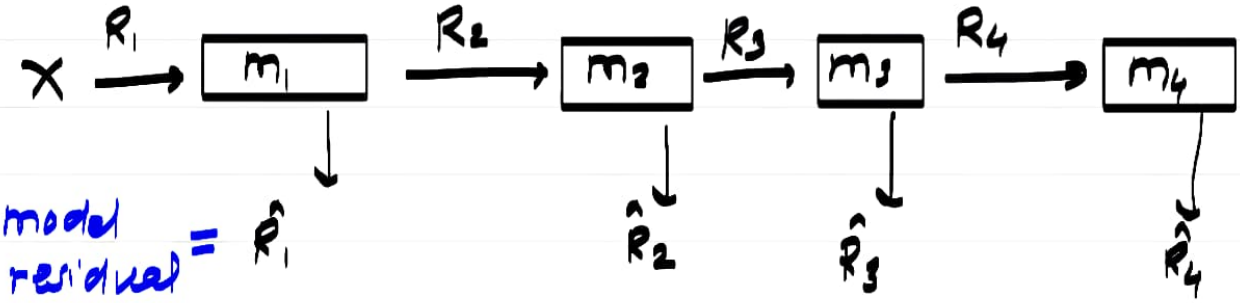
- $\hat{y}_i$: preds after model m_i
- $\hat{y}_{i-1}$: preds from previous model
- $\hat{R}_i$: residual preds from model m_i

7) Repeat from step 3 to step 6 untill the stopping criteria is met

* Stopping criteria for GBM

- Decided number of weak learners are built
- Residual values have become small and are not changing much by adding extra models

Final model prediction after training:



$$\hat{y}_1 = \hat{y}_0 + \alpha \hat{R}_1$$

$$\hat{y}_2 = \hat{y}_1 + \alpha \hat{R}_2 = \hat{y}_0 + \alpha \hat{R}_1 + \alpha \hat{R}_2$$

$$\hat{y}_3 = \hat{y}_2 + \alpha \hat{R}_3 = \hat{y}_0 + \alpha (\hat{R}_1 + \hat{R}_2 + \hat{R}_3)$$

$$\hat{y}_4 = \hat{y}_3 + \alpha \hat{R}_4 = \hat{y}_0 + \underbrace{\alpha (\hat{R}_1 + \hat{R}_2 + \hat{R}_3 + \hat{R}_4)}_{\text{aggregation}}$$

final GBM prediction = $\hat{y}_4$